

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

ANSWER KEY AND MARKING SCHEME

Biology: Class 9 (Science Group)

Syllabus: Chapter 1 (The Science of Biology) | Chapter 2 (Biodiversity) | Chapter 3 (The Cell) | Chapter 4 (Cell Cycle) | Chapter 5 (Tissues, Organs and Organ Systems) | Chapter 6 (Molecular Biology) | Chapter 7 (Metabolism, complete)

Total Marks: 38

Time: 1 Hour 45 Minutes

Paper: Exam-17 (Version B)

Answer letters: 1-C 2-D 3-A 4-B 5-B 6-A 7-A 8-C 9-D 10-B (A 3, B 3, C 2, D 2)

SECTION A: MCQ Key

#	Ans	Why (and why the others are wrong)	Source in her textbook
1	C observation	Of the options given, observation comes first: the biological method starts with recognition of the problem, then observation, then a hypothesis, deductions and experiments.	Ch 1, 1.6 Biological Method (p.11 to 12); Exercise Section I MCQ 7 (p.17); SLO 8
2	D data	The results recorded from an experiment are called data. Data are then analysed to accept or reject the hypothesis.	Ch 1, 1.6 Biological Method step 6 (p.12); Exercise Section I MCQ 8 (p.17); SLO 8
3	A phylum	Taxonomic ranks: kingdom, phylum, class, order, family, genus, species. Organisms in the same class are always in the same phylum (the rank above), but not necessarily in the same order, family or genus.	Ch 2, 2.5 Taxonomic Ranks of Classification (p.25); Exercise Section I MCQ 17 (p.29); SLO 5
4	B intergrana	Book: some thylakoids of adjacent grana fuse to form intergrana. Cisternae are found in the ER and Golgi apparatus, cristae in mitochondria, and the tonoplast is the vacuole membrane.	Ch 3, 3.1.1 Plastids: Chloroplast (p.36); SLO 3
5	B lipids	Vitamins A, D, E and K are lipid in nature; vitamin D is made from cholesterol (a steroid).	Ch 6, 6.4.2 Functions of Lipids point 7 (p.82), Table 6.1 (p.79); Exercise Section I MCQ 10 (p.92); SLO 4
6	A brain	ATP usage in body: the brain uses maximum energy in the form of ATP, i.e. 25% of ATP.	Ch 7, 7.8.3 ATP usage in body, below Fig. 7.12 (p.107); SLO 6 and SLO 8
7	A do not depend directly on light	Book: these reactions do not depend directly on light, that is why it is called dark reactions or light independent reactions. They take place in the stroma and do not release oxygen.	Ch 7, 7.7.1 (p.102) and 7.7.1.2 Light Independent Reactions (p.104); Exercise Section II Q13 (p.111); SLO 7
8	C two ATP molecules	Book: electrons from each NADH give enough energy for three ATP, however each FADH ₂ produces only two ATP molecules.	Ch 7, 7.8.3, 4. Electron Transport Chain (p.107); SLO 8
9	D wine making and baking	Yeast carries out alcoholic fermentation (ethyl alcohol + CO ₂), used in wine making and baking. Yogurt and cheese are made by bacteria; muscles make lactic acid.	Ch 7, 7.8.1.1 Importance of Anaerobic Respiration (p.106); SLO 8
10	B water	Oxygen is the final electron acceptor and it ultimately combines with hydrogen ions to produce water.	Ch 7, 7.8.3, 4. Electron Transport Chain (p.107); SLO 8

SECTION B: Short Questions (model answers)

2(i) (4 marks)

(a) $6\text{CO}_2 + 12\text{H}_2\text{O} \xrightarrow{\text{light, chlorophyll}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O} + 6\text{O}_2$ **1** (light and chlorophyll must be shown; also accept the Ch 6 p.83 form $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ with light and chlorophyll)

(b) Pyruvic Acid $\rightarrow$ Ethyl Alcohol + Carbon dioxide **1**

(c) $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{Energy (ATP)}$ **1**

(d) Lactic acid (Pyruvic Acid $\rightarrow$ Lactic Acid). **1**

Source: Ch 7, Exercise Section II Q11 and Q12 (p.111); 7.7 (p.102); 7.8.1 (p.105); 7.8.2 (p.106); SLO 7, 8.

2(ii) (4 marks)

(a) Chloroplasts (they contain chlorophylls; the other coloured plastids, chromoplasts, are coloured other than green). **1**

(b) Chromoplasts (coloured other than green: they may be red, pink, yellow, blue, purple etc.). **1**

(c) In flower petals, where they **attract insects** that help in **pollination**. **1**

(d) Chromoplasts in the walls of ripened fruits **attract birds and animals** that help in **seed dispersal**. **1**

Source: Ch 3, 3.1.1 Plastids: Chloroplast, Chromoplast (p.36); Exercise Section II Q10 and Section III Q5 (p.47); SLO 3.

2(iii) (4 marks)

Any four, 1 mark each:

Aerobic respiration	Anaerobic respiration
Takes place in the presence of oxygen	Takes place in the absence of oxygen
Glucose is completely broken down	Glucose is not completely oxidized
End products are carbon dioxide and water	End products are ethyl alcohol and carbon dioxide (yeast) or lactic acid (muscles)
Comparatively large amount of energy is produced	Comparatively less energy is produced
Four phases: glycolysis, formation of acetyl CoA, Krebs cycle, ETC	Glycolysis, then pyruvic acid is converted into ethyl alcohol and carbon dioxide, or into lactic acid

*Source: Ch 7, 7.8.1 and 7.8.2, 7.8.3 (p.105 to 106); SLO 8.***SECTION C: Long Questions (model answers)****3(i) (8 marks)**

(a) Sketch (3 marks), like Fig. 7.10: light falling on **photosystem II**; electron pair going to the **primary electron acceptor**; **water splitting** into O_2 and H^+ ; electrons passing along the **electron transport chain** with **ATP** formed; light on **photosystem I**; **$NADP^+ \rightarrow NADPH$** . Neat sketch with arrows **1**; at least six correct labels **2 (three to five labels: 1)**.

(b) Description (5 marks, 1 per step):

- The reactions take place in the **thylakoid membranes**, where pigments are arranged in **photosystems** (I and II); the process starts from photosystem II. **Chlorophyll a of photosystem II absorbs light** and emits a pair of electrons, accepted by the **primary electron acceptor**. **1**
- At the same time a water molecule splits into an oxygen atom, two H^+ and two electrons. This makes up the electron loss of photosystem II and is called **photolysis of water**. **1**
- Oxygen** is released out of the leaf through **stomata**. **1**
- The electron pair passes through the **electron transport chain (ETC)**, gradually releasing energy used for **ATP synthesis**. **1**
- Light also acts on **photosystem I**, which gives out an electron pair; these electrons and two H^+ of water **reduce $NADP^+$ to $NADPH$** . **1**

*Source: Ch 7, 7.7.1.1 Light Dependent Reactions and Fig. 7.10 (p.103 to 104); Exercise Section III Q6 (p.111); SLO 7.***3(ii) (8 marks)**

(a) Four phases, 1 mark each:

Prophase II: chromatin coils into chromosomes, the spindle apparatus is organised, and the nuclear envelope disintegrates. **1**

Metaphase II: chromosomes line up separately at the equator, each attached by spindle fibres from both poles. **1**

Anaphase II: centromeres divide and the chromatids are pulled to opposite poles. **1**

Telophase II: chromatids reach the poles and uncoil, and nuclear envelopes form; cytokinesis gives **four haploid daughter cells**. **1**

(b) Similarities, any two: divided into the **same four phases** (prophase, metaphase, anaphase, telophase); chromosomes **line up separately** at the equator, each attached by a fibre **from both poles** (like mitosis); **centromeres divide** and chromatids move to opposite poles. **2**

(c) Differences, any two: meiosis II starts with **haploid** cells, mitosis with diploid cells; **no DNA replication** before meiosis II (interphase II), but the genome is replicated before mitosis; meiosis II products are **not genetically identical** (because of crossing over in meiosis I), mitosis gives genetically identical cells; meiosis II occurs in germ cells to form gametes/spores, mitosis in somatic cells. **2**

Source: Ch 4, 4.5 Meiosis: Interphase II, Meiosis II (p.55 to 56); 4.3 Mitosis (p.50 to 51); comparison table (p.57); Exercise Section III Q3 (p.63); SLO 2, 3.

End of Answer Key